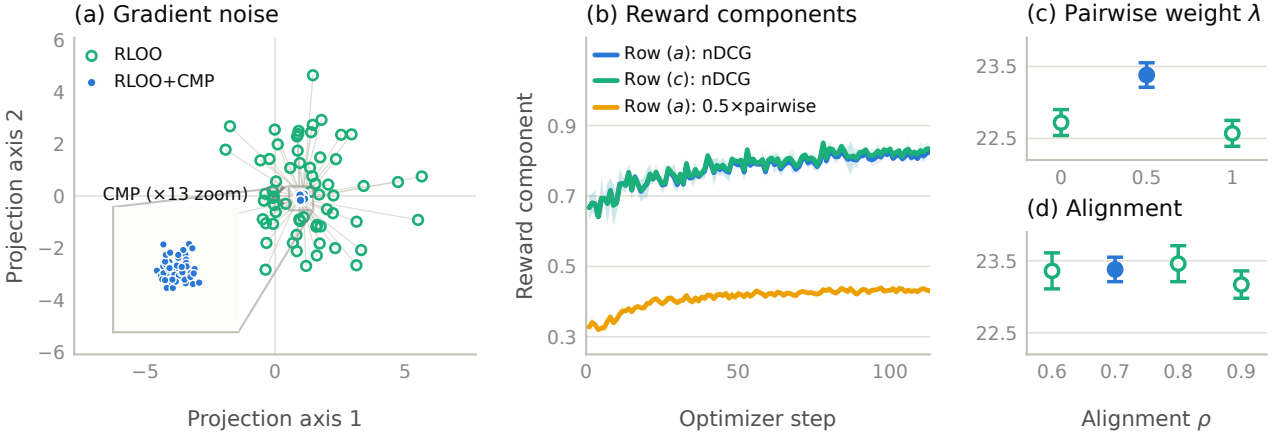
(a) Gradient noise
RLOO
RLOO+CMP
CMP (×13 zoom)
Projection axis 2
Projection axis 1
6
4
2
0
−2
−4
−6
−5
0
5
(b) Reward components
Row (a): nDCG
Row (c): nDCG
Row (a): 0.5×pairwise
Reward component
0.9
0.7
0.5
0.3
0
50
100
Optimizer step
(c) Pairwise weight λ
23.5
22.5
0
0.5
1
(d) Alignment
23.5
22.5
0.6
0.7
0.8
0.9
Alignment ρ